

Merkle Science: Safeguarding the crypto ecosystem with predictive intelligence

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ABOUT MERKLE SCIENCE

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About Merkle Science

Merkle Science is a cryptocurrency risk and intelligence platform creating the infrastructure to enable crypto's safe and healthy growth. Its team is building the technology that aims to set the standard for the next generation of financial safeguards.

Industries: Technology

Location: Singapore

Google Cloud results

- Frees team to focus on application building by increasing developer bandwidth by 10%
- Enables ultra-fast transaction screening by enabling latencies of under 2 seconds 99% of the time
- Supports developer team agility with proactive training on infrastructure migration to Google Kubernetes Engine
- Opens up investment for product

Released by

searche°

About Searche

Searche is one of the largest Google Cloud partners with over 12 years experience and more than 150 Google Cloud certifications.

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Google Cloud (<https://cloud.google.com/>)

App Engine (<https://cloud.google.com/appengine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Functions
(<https://cloud.google.com/functions/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Virtual Private Cloud (<https://cloud.google.com/vpc>)

development
by saving up
to 20% in
cloud costs

Nearly a billion people are projected to be cryptocurrency users by 2028, with revenue in the industry growing at a compound annual growth rate (CAGR) of 8.62 percent to reach \$71.7 billion (<https://www.statista.com/outlook/fmo/digital-assets/cryptocurrencies/worldwide>) by that year.

Amid such a remarkable rise in crypto interest there is another fast growing trend in the space, which is fraud. Total illicit transaction value more than quadrupled (<https://globalinvestigationsreview.com/review/the-asia-pacific-investigations-review/2024/article/emerging-trends-in-crypto-fraud>) from \$4.6 billion in 2018 to \$20.6 billion in 2022.

Merkle Science (<https://www.merklescience.com/>) is a US-based startup on a mission to make the crypto world safe and secure through advanced predictive intelligence. Founded in 2018, Merkle Science empowers crypto companies, financial institutions, and government entities to detect, investigate, and mitigate illegal activities involving cryptocurrencies.

Its proprietary Behavioral Rule Engine analyzes unusual patterns of crypto trading data to enable compliance teams to detect potentially illicit transactions in real time. It also features advanced attribution of wallets to help fulfill know-your-

customer (KYC) and anti-money laundering (AML) obligations.

To succeed, Merkle Science requires a nimble, autoscaling computing environment to meet any spike in screening requests from any corner of the crypto world. It also needs powerful data analytics to power its predictive intelligence.

Originally with one of the major public cloud providers, Merkle Science found itself hampered by the need for manual configurations and a hard-to-use user interface, and decided to migrate to Google Cloud. It was attracted by a combination of Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE), for the power and simplicity of a fully automated microservices environment, and BigQuery (<https://cloud.google.com/bigquery>) as an autoscaling data warehouse to enable it to crunch unlimited data from blockchains.

"Google Kubernetes Engine and BigQuery form the backbone to our teams," says Lakshmi Narasimham, Director of Product Engineering, Merkle Science.

"Together they are the gold standard in their fields, bringing advantages such as unlimited scale, power and agility, and they are enabling us to chart the future of crypto risk management."

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A proactive partner to empower teams to optimize cloud solutions

One of the key considerations that went into tapping Google Cloud was the way it provided proactive support that went beyond a client-vendor relationship to be "a true partnership," both with the Google Cloud team itself, and Google Cloud approved cloud partner [Searce](https://www.searce.com/) (<https://www.searce.com/>).

"We truly feel that Google Cloud has empathy for customers. Google Cloud understood our pain points and worked with us to solve our problems" says Lakshmi Narasimham, Director of Product Engineering, Merkle Science. "With Google Cloud we have a partner who is constantly listening to us, either directly or through the Searce team. Our previous provider, by contrast, was a completely passive vendor."

Searce has been involved in the partnership since 2020 from the days when Merkle Science has been utilizing Google Cloud credits. Searce started with an infrastructure review for Merkle Science and then did a modernization cost to optimize their architecture. Since then, Searce has driven multiple engagements for modernization of the Merkle Science DevOps pipeline.

When Merkle Science began its Google Cloud journey, it deployed [App Engine](https://cloud.google.com/appengine) (<https://cloud.google.com/appengine>) for its infrastructure needs, and found itself slowed down

by manual infrastructure configurations, an issue it had also experienced with its original provider.

Seeking guidance from Searce, Merkle Science was able to automate App Engine processes that freed the development team from mundane chores and enabled it to focus on creative application development. Specifically, the Searce team helped Merkle Science in providing the automation script for existing App Engine infrastructure, helping the startup manage multiple production environments.

Overall, the Searce Infrastructure Assessment helped Merkle Science assess and finalize their DevOps strategy which resulted in initiatives to automate their manual DevOps processes leveraging Infrastructure as a Code (IaaC).

"It was a smooth journey with Searce," says Karambir Singh Nain, lead backend engineer at Merkle Science. "With their help, we were able to serve our businesses with less effort, because we could focus on our product development rather than dealing with infrastructure related issues."

Searce then advised Merkle Science on transferring its computing infrastructure to GKE for a more agile, autoscaling microservices-based architecture, creating a proof-of-concept and giving training to the Merkle Science team to migrate its entire backend application infrastructure to GKE. The idea was to provide the team hands-on expertise in moving the infrastructure to GKE to gain more fine-tuned insight into its advantages.

The ease of use and automation of GKE is proving critical to enabling Merkle Science's small team of developers and engineers, in particular driving smoother and faster releases. The time required for product releases has now been reduced from roughly 30 minutes to less than two minutes. In addition, GKE is invaluable because it is fully compatible with clients who work in environments of other major public clouds.

"With Google Kubernetes Engine, we now have a more scalable architecture that enables us to meet any demand in screening requests for transactions," says Nain. "And with a very small team, we are able to scale up clusters, instead of going into complex and time-consuming configurations, saving us 10 percent of our developer bandwidth."

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Merkle Science

Deploying a robust ecosystem for ETL Optimization with BigQuery

For Extract, Transform, Load (ETL) operations, Merkle Science deploys a combination of Cloud Functions (<https://cloud.google.com/functions>) and Cloud Run (<https://cloud.google.com/run>), alongside Cloud SQL (<https://cloud.google.com/sql>) as its relational database, to enable the powerful data streams that are funneled into BigQuery for machine learning-enabled data analytics to detect fraud.

It's a nimble system that circumvents the need for manual data handling, saving the team time and reducing costs of ingress and egress, due to the solutions being part of the same ecosystem.

Overall, says Narashimham, the system enables the startup to unleash the transformative power of BigQuery. "In our data layer, we are very data heavy as lots of streams are coming from blockchains. We

have to manipulate, store, and analyze them, and BigQuery is optimal for that," he says. "Unlike any other data warehousing solution, BigQuery enables the powerful, scalable data pattern detection we need to fulfill our obligations to customers."

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Unlocking powerful measurable results through solution optimization

The results of the comprehensive migration to an architecture built around GKE and BigQuery have been significant. Saved from manual provisioning tasks, Merkle Science estimates it has saved more than 1,000 people hours since migration to GKE, time it can devote to development of core products.

Merkle Science deploys Cloud Load Balancing (<https://cloud.google.com/load-balancing>) for secure low latency global connections, distributing loads across multiple regions to services within its Virtual Private Cloud (<https://cloud.google.com/vpc>), for maximum speed and security. The startup had a goal of improving latencies for screening crypto addresses to under two seconds 99 percent of the time, a significant improvement from the previous 75 percent performance. It easily achieved the objective thanks to the power of Cloud Load Balancing.

Moreover, the firm has achieved operational savings of up to 20 percent on its cloud costs, thanks in large part to the guidance provided by the Google Cloud and Searce teams on optimizing use of Google Cloud solutions. This represents funds that the firm can plough into investment in creative solution building.

"The Google Cloud and Searce teams are actively advising us on how to bring down costs in our cloud computing spend, which we find truly remarkable," says Nain. "It's a testament to the partnership ethos we enjoy with both of them."



Transforming the future of crypto security with AI solutions

Looking toward the future, Merkle Science anticipates deployment of generative AI solutions from Google for numerous potential use cases. One is the automation of request for proposals (RFPs) from clients that tend to be a long and cumbersome process when handled manually.

Another area where the startup sees deploying AI is improving developer productivity for the testing of projects. And it foresees potential for improving its internal behavior engines for risk modeling, possibly deploying Google Cloud AI tools for this task.

In all of these cases, Merkle Science is studying the potential adoption of Google Cloud AI solutions such as Vertex AI (<https://cloud.google.com/vertex-ai>), alongside Gemini deployment. In the partnership spirit, it is seeking training workshops with the Google Cloud team to discover applications for specific use cases.

"There's an exciting future we foresee in partnership with Google Cloud, developing solutions for the crypto world," says Narasimham. "We look forward to the opportunity to explore AI products and services to help drive the next stages of our evolution."